Experiment Report: Push-Up k=2 Extension (+10K)

NL_Hecate — Nested Learning Push-Up Pipeline

Run ID: `baseline_pushup_p2_k2_ext10k` — Date: 2026-03-15

GPU: A6000 (GPU 0) — Duration: 2.3 h — 609 tok/s

Outcome: CONFIRMED FAILURE — Push-up k=2 initialization is ruled out

Purpose

The push-up k=2 experiment (report 2) showed regression across all metrics. This extension tests whether **additional training time** resolves the problem. 10K more steps on fresh data are added to the k=2 push-up's step-20K checkpoint.

Hypothesis: Given sufficient joint training, L0 and L1 will eventually differentiate and the model will recover from the initial regression.

Answer: No. The levels remain trapped in a degenerate basin. The extension confirms that the push-up initialization method is fundamentally flawed, not undertrained.

Control: A concurrent fresh k=2 from random init ran on GPU 1 for direct comparison. Results from this run (at step 10K) are included as contrast throughout.

Configuration

Table 1: Extension run configuration. Only differences from the original push-up k=2 are shown.

Parameter	Value	Change from p2_k2
Steps	10,000 (20K→30K total)	+10K extension
Donor checkpoint	`baseline_pushup_p2_k2/model.safetensors`	k=2 at 20K
Warmup steps	250	Reduced (was 500)
Data seek	17,920,512	100% fresh data
LR	3×10^{-4} cosine (re-started)	New cosine cycle

All other parameters identical: d=512, 4 blocks, k=2, chunk_sizes=[1,8], m_norm_clamp disabled, TNT hierarchical, dolmino_100b.

Results Summary

Table 2: Key metrics at milestones (steps 20K–30K).

Step	Loss	PPL	Grad Norm	Block CV
20,000	3.935	51	2.10	0.328
22,000	3.935	51	2.68	0.371
24,000	5.582	266	3.84	0.269
26,000	2.838	17	2.03	0.303
28,000	3.214	25	2.09	0.296
30,000	2.906	18	2.82	0.329

Table 3: Smoothed loss trajectory (50-step windows).

Window	Mean Loss	Mean Grad Norm
Steps 20K–21K	3.458	2.254
Steps 22K–23K	3.444	2.403
Steps 24K–25K	3.312	2.193
Steps 26K–27K	3.391	2.234
Steps 28K–29K	3.297	2.367
Steps 29K–30K	3.276	2.255

Summary: Loss improved marginally (3.46 → 3.28 mean), still well above the k=1 baseline (2.84) and the fresh k=2 trajectory (3.70 at step 10K, trending down). The 10K extension bought almost nothing.

Loss: Full Experimental Context

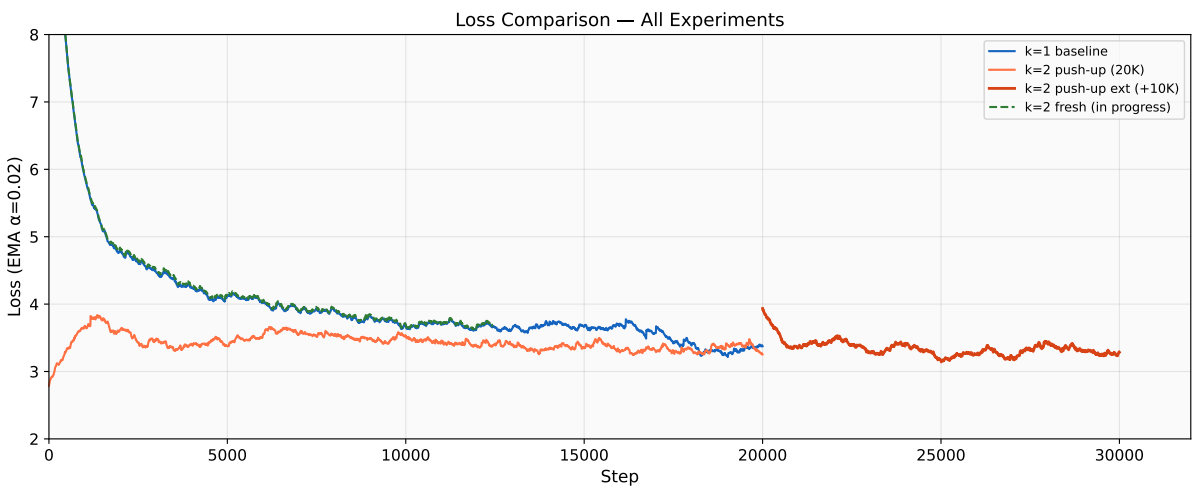

Figure 1: Loss comparison across all four experiments. The extension (dark red) continues the push-up's plateau from steps 20K–30K. The fresh k=2 (green dashed) is already converging faster from random init.

The four-experiment overlay tells the complete story:

- **k=1 baseline** (blue): Converges to 2.84 at step 20K.
- **Push-up k=2** (orange): Regresses from 2.79, plateaus at ∼3.4.
- **Push-up ext +10K** (dark red): Continues the same plateau, ending at ∼3.28.
- **Fresh k=2** (green): Starts at 11.0, reaches 3.70 at step 10K with a steeper descent slope. Projected to reach ∼3.0 by step 20K.

The push-up spent 30,000 total steps (20K + 10K extension) to reach a loss of 3.28. The fresh k=2 will likely beat that in under 15K steps from random init. More training cannot fix a broken initialization.

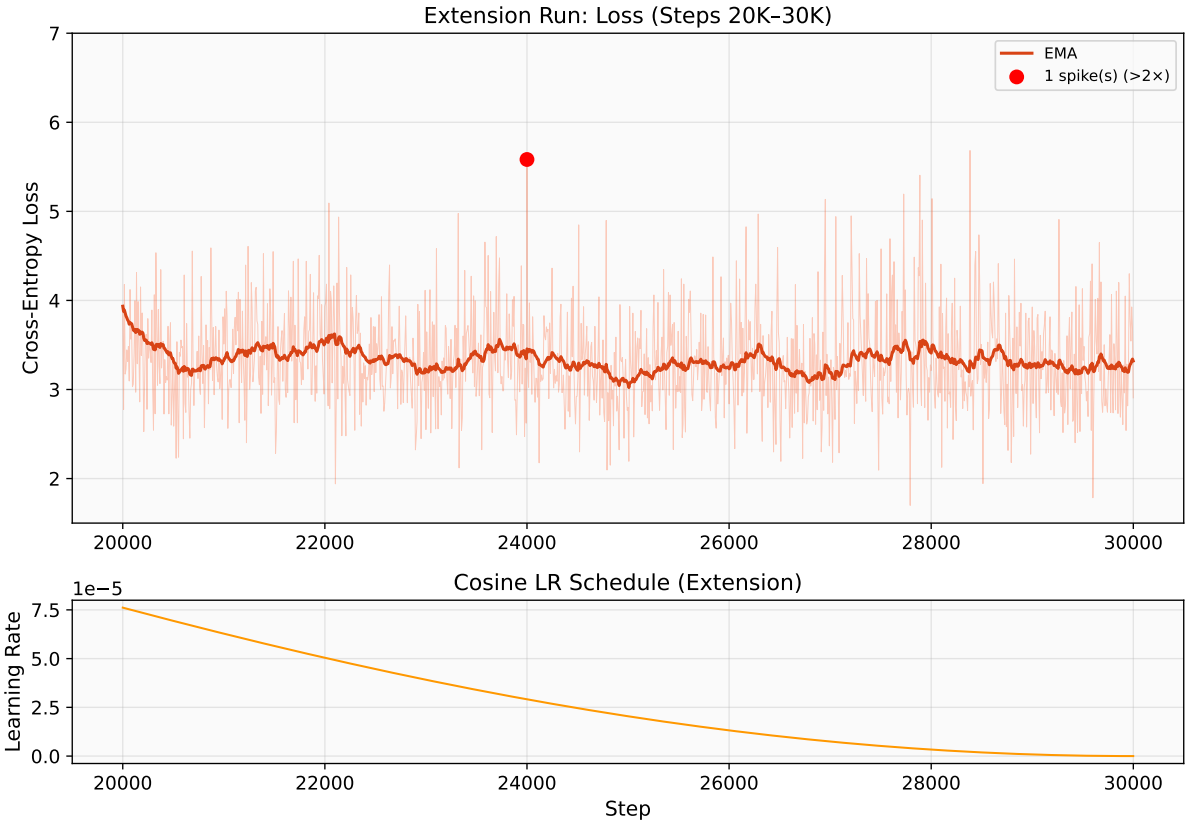

Figure 2: Extension run loss detail. One major spike at step 24K (loss=5.58) correlates with L1 theta instability (see Section 6).

Gradient Analysis

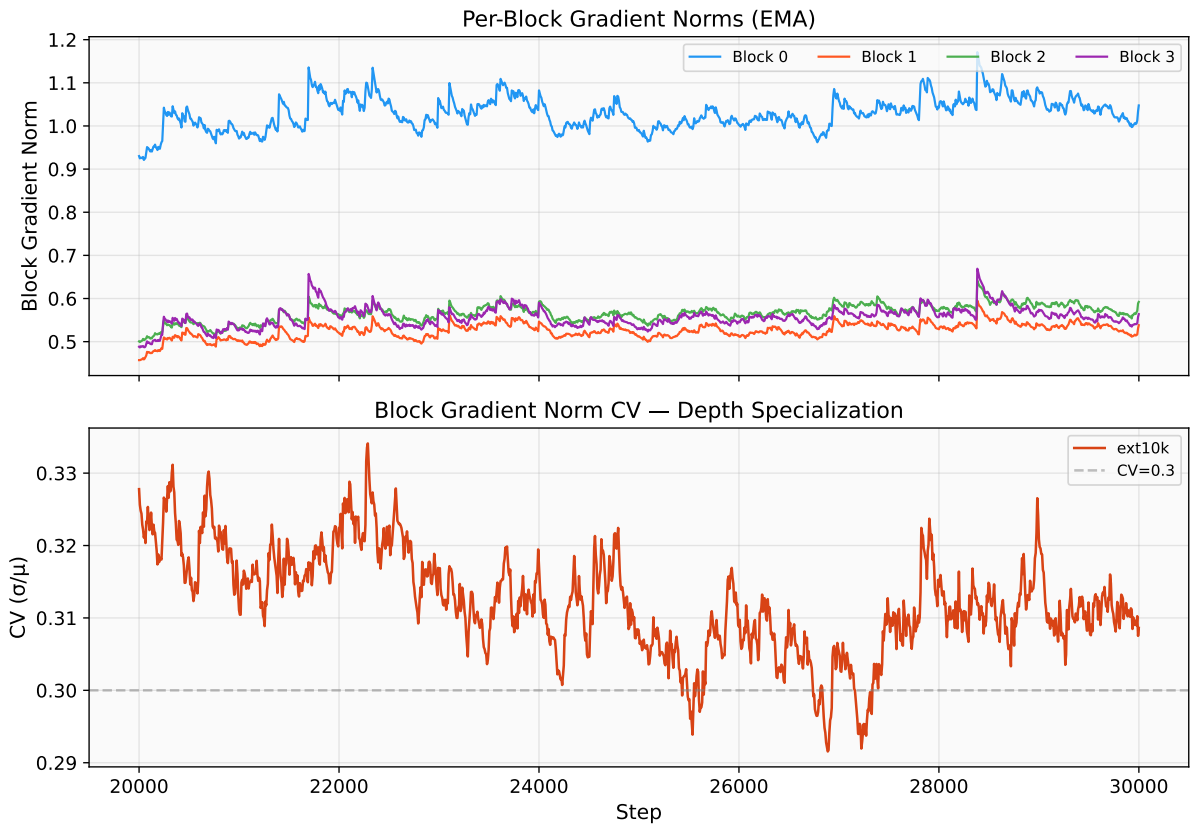

Figure 3: Per-block gradient norms and CV during the extension. CV hovers at ∼0.30, below the 0.3 threshold for meaningful specialization.

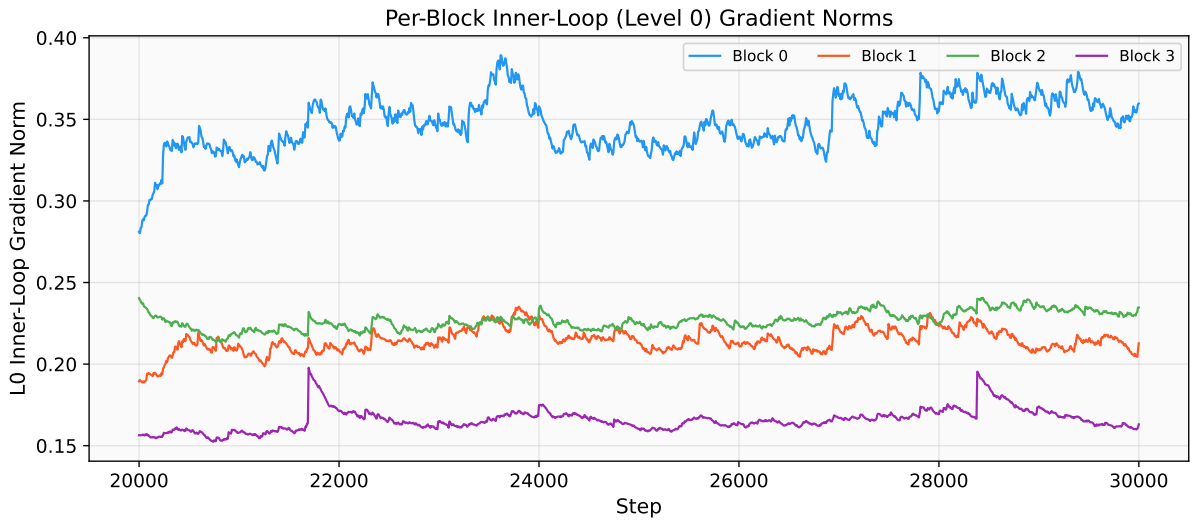

Figure 4: L0 inner-loop gradient norms. Block 0 dominates (∼0.35) but no growth trend is visible — the memory mechanism has stagnated.

Gate Diagnostics: The Oscillation Problem

This section presents the tape-level diagnostics that explain **why** the push-up fails. The extension run logged 20 tape snapshots with per-level gate statistics (alpha, theta, eta) for every block. A concurrent fresh k=2 run provides the control comparison.

6.1 Theta (Inner-Loop Learning Rate): Oscillation vs Convergence

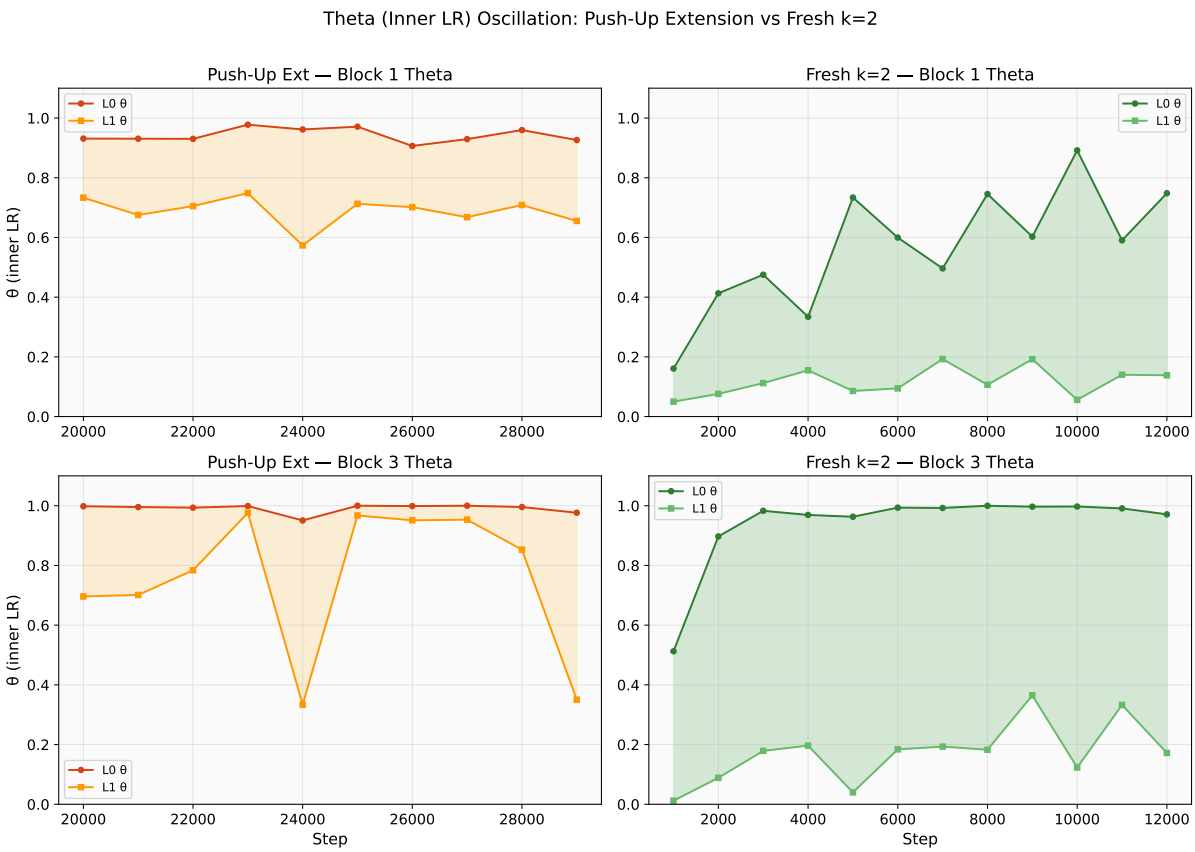

Figure 5: Theta (inner-loop LR) for L0 and L1 across two blocks. **Left:** Push-up extension — L1 theta oscillates wildly, especially in Block 3. **Right:** Fresh k=2 — L0 rises to ∼0.9, L1 stays low (<0.2). Shaded area shows L0-L1 gap.

Table 4: Block 3 L1 theta: push-up extension showing extreme oscillation.

Step	L0 θ	L1 θ	Behavior
20,000	0.998	0.696	Moderate
23,000	0.999	**0.976**	Rockets to ceiling
24,000	0.951	**0.334**	Crashes (loss spike)
25,000	1.000	**0.967**	Rockets back up
28,000	0.996	0.853	Partial recovery
29,000	0.977	**0.351**	Crashes again

L1 theta in Block 3 exhibits **0.6+ swings** between adjacent tape snapshots. It oscillates between "maximally aggressive" (∼0.97) and "nearly shut off" (∼0.33) with no convergence

over 10K steps. L0 theta, inherited from k=1 training, remains pinned at the ceiling (>0.95) with no room to modulate.

In contrast, the fresh k=2 shows clean separation: L0 rises to ∼0.99 while L1 stays below 0.2 throughout. Each level has found a distinct, stable role.

The step-24K loss spike (5.58) directly correlates with L1 theta crashing to 0.334. When L1 abruptly changes its inner-loop learning rate by 3×, the forward pass geometry shifts and the model produces incoherent outputs.

6.2 Alpha (Retention): No Timescale Separation

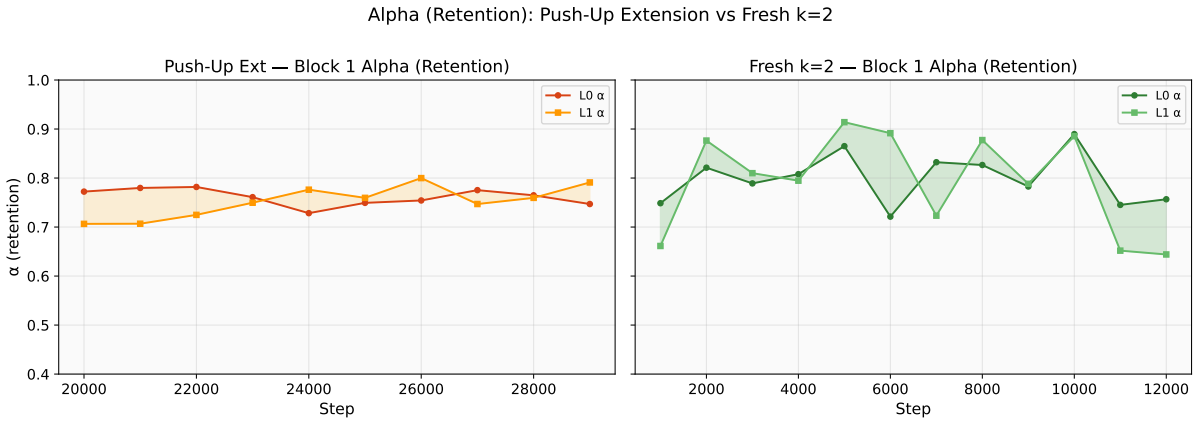

Figure 6: Alpha (retention) for Block 1. **Left:** Push-up extension — L0 and L1 alpha overlap with no separation trend. **Right:** Fresh k=2 — alpha values cross and wander but show slightly more range.

Alpha presents the weakest differentiation signal in *both* experiments:

- **Push-up ext:** L0 alpha ≈ 0.75, L1 alpha ≈ 0.75. Delta oscillates ±0.05 with no trend. The levels have identical retention — no memory timescale separation.
- **Fresh k=2:** L0 alpha ≈ 0.80, L1 alpha ≈ 0.82. Slightly wider range but still largely overlapping. Occasional excursions to −0.17 delta suggest the potential for separation, but it has not stabilized by step 10K.

This motivates the retention initialization experiment: if alpha=0.8 produces minimal separation, starting at alpha=0.5 (maximum freedom to move in either direction) may allow the model to discover distinct retention rates per level.

6.3 Eta (Output Gate): Moderate Separation

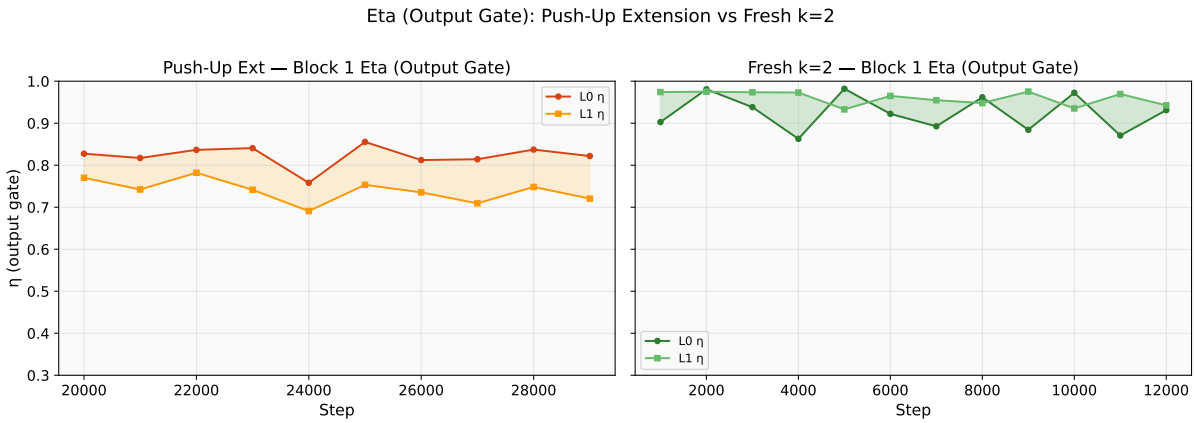

Figure 7: Eta (output gate) for Block 1. Both experiments show L0 eta > L1 eta, with the fresh k=2 exhibiting wider separation.

Eta shows the most consistent differentiation of the three gates:

- **Push-up ext:** L0 eta ≈ 0.83, L1 eta ≈ 0.74. Gap ∼0.09.
- **Fresh k=2:** L0 eta starts higher and L1 lower, with gap ∼0.10–0.15.

L0 contributes more to the output than L1 in both experiments, which is consistent with L0 firing every token (more data, more confident) while L1 fires every 8th.

6.4 DGD Update Magnitude Ratio

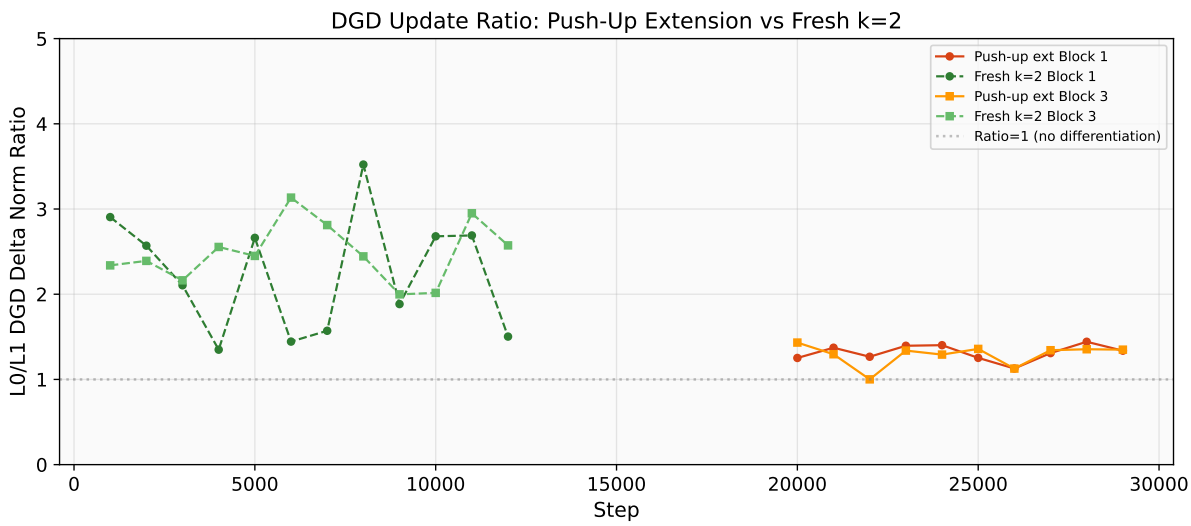

Figure 8: L0/L1 DGD delta norm ratio. Push-up ext (circles) is stuck at ∼1.3. Fresh k=2 (squares) reaches 2.5–3.5 — L0 makes much larger memory updates.

The DGD ratio measures how much larger L0's memory updates are compared to L1's.

- **Push-up ext:** Stuck at 1.1–1.4 for the entire 10K extension. No growth.
- **Fresh k=2:** Ranges from 1.4 to 3.5, with higher values as training progresses. L0 is making 2–3× larger updates, consistent with its 8× higher firing rate.

A healthy DGD ratio >2 means the levels are making updates of different magnitudes, which is a necessary condition for them to learn distinct representations.

6.5 Output Gradient Norms: Identical in Both Experiments

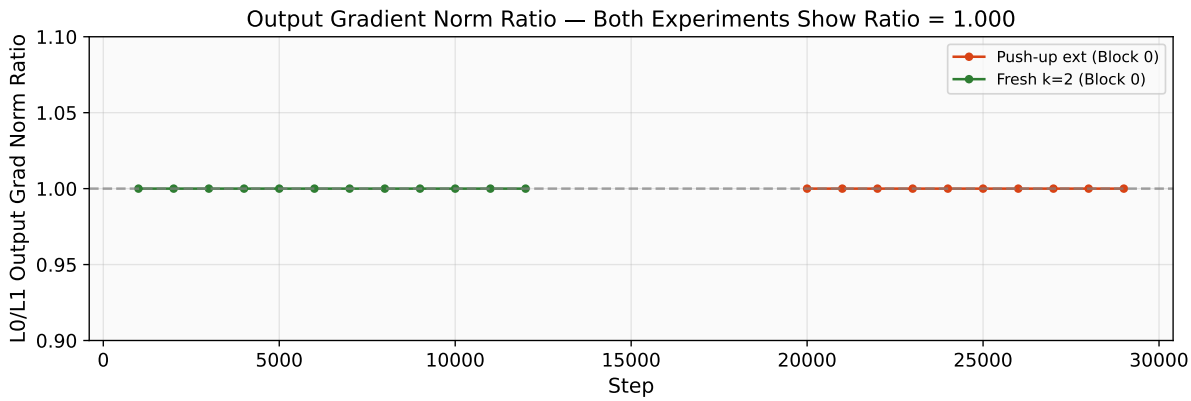

Figure 9: L0/L1 output gradient norm ratio = 1.000 in both experiments. This is a structural property of MAG composition, not a bug.

The output gradient norm ratio is **exactly 1.000** across all snapshots in both experiments. L0 and L1 receive identical gradient signals from the loss. This is a structural property of the MAG (memory-attention-gated) composition: both levels contribute to a single gated sum, so the backward pass distributes the same gradient to both.

Implication: Level differentiation cannot come from the output gradient. It must come entirely from:

1. The **frequency gap** (8×) — L0 fires every token, L1 every 8th
2. The **inner-loop gate gradients** — alpha, theta, eta are per-level learnable
3. The **DGD update magnitude** — different because of accumulated inner-loop state

The fresh k=2 achieves differentiation through pathways 1–3 despite identical output gradients. The push-up cannot because the cloned initialization neutralizes all three pathways.

Training Throughput

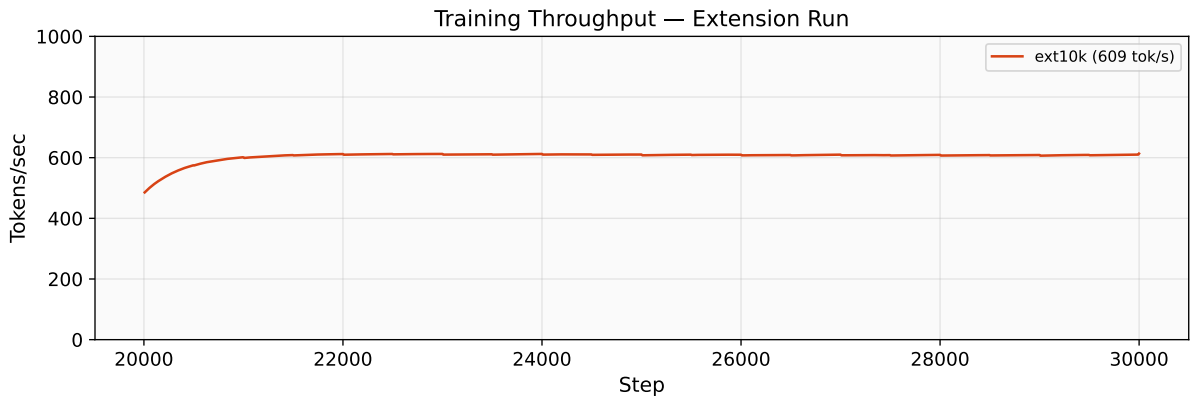

Figure 10: Extension run throughput: stable at 609 tok/s.

Root Cause: Cloned Initialization Traps Both Levels

The extension confirms the diagnosis from the push-up report (report 2), with additional gate-level evidence:

1. **L0 theta saturated at ceiling** (>0.95) — inherited from k=1, cannot adapt.
2. **L1 theta oscillating** (0.33 ↔ 0.97) — no stable equilibrium exists because L1 receives the same gradient as L0.
3. **Alpha undifferentiated** ($\Delta \approx \pm 0.05$) — no memory timescale separation. Both levels retain at the same rate.
4. **DGD ratio stuck at ∼1.3** — both levels make nearly identical magnitude updates, preventing representation divergence.
5. **Output gradients identical** (ratio = 1.000) — differentiation pressure comes only from frequency gap and inner-loop gates.
6. **10K more steps did not help** — this is a basin problem, not a budget problem.

The push-up clones a converged single-level model into two copies. Both start in the same optimization basin with identical weights, identical gates, and identical gradients. The 8× frequency gap is insufficient pressure to break this symmetry when the initialization is degenerate.

Control: Fresh k=2 (Concurrent Run)

Table 5: Push-up ext (step 27K) vs fresh k=2 (step 10K) gate diagnostics.

Metric	Push-Up Ext	Fresh k=2	Verdict
L0-L1 theta delta (Blk 1)	+0.20 to +0.27	**+0.64 to +0.84**	Fresh 3× wider
L1 theta stability (Blk 3)	0.33–0.98	**0.01–0.36**	Fresh stable
L0-L1 alpha delta	±0.05	±0.17	Both weak
DGD ratio (Blk 1)	1.1–1.4	**1.4–3.5**	Fresh 2× higher
Block CV	0.27–0.37	**0.40–0.58**	Fresh more specialized
Loss spikes (>2×)	1 in 10K	**0 in 10K**	Fresh stable

The fresh k=2 achieved stronger differentiation in **2K steps** than the push-up achieved in **30K steps**. Every gate diagnostic favors fresh initialization.

Conclusions

10.1 Verdict

The k=1 → k=2 push-up initialization method is **ruled out**. The 10K extension confirmed that additional training does not resolve the differentiation failure. The levels are trapped in the same optimization basin and cannot escape.

10.2 Key Findings

1. **Theta oscillation is the proximal failure mode.** L1 theta bounces 0.6+ between snapshots because it receives the same gradient as L0 but is trying to find a different role. The oscillation directly causes loss spikes.
2. **Alpha never separated.** Neither experiment produced strong retention-rate differentiation at alpha=0.8. This is the motivation for the retention=0.5 experiment: test whether alpha is learnable from a neutral starting point.
3. **Output gradient identity is structural.** Both levels will always receive the same output gradient under MAG composition. Differentiation must come from frequency, gates, and inner-loop dynamics alone.
4. **Fresh initialization works.** The concurrent fresh k=2 proves that two levels *can* self-organize from random init. The 8× frequency gap provides sufficient structural pressure when the initialization is not degenerate.

10.3 Implications for the Push-Up Pipeline

The push-up strategy is not invalidated — it is **reframed**. The correct approach builds from the foundation up:

1. **Fresh** k=2: Train the two slowest levels (L2, L3) from random init until differentiated and converged. These become the anchor.
2. **Push-up two levels**: Add L0 and L1 simultaneously on top of the established L2/L3. The new levels are a matched pair with the same 8× ratio that already proved viable in the fresh k=2 run. L1 meshes with L2 at 8×; L0 meshes with L1 at 8×. The gear train self-assembles.
3. **Retention initialization**: Use the results of the retention=0.5 experiment to determine whether new levels should start with neutral retention (0.5) or need a floor until the inter-level relationship stabilizes.

10.4 Next Experiments

1. **Complete fresh** k=2 **(GPU 1)** — confirm differentiation holds to 20K.
2. **Retention=0.5 probe (10K)** — does the model set its own retention rate?
3. **If successful, extend to 50K** — two comparison points (0.5 vs 0.8 retention).
4. **Two-level push-up** k=2 → k=4 — the real test of the gear-train hypothesis, using the best k=2 donor.